Preliminary Data Sheet

SV6166F

Low Power WiFi Single Band 802.11 b/g/n SoC for IoT

General Description

The SV6166F is a low-power single chip device providing for the highest level of integration for internet of thing embedded systems. It is designed to support all mandatory IEEE 802.11b data rates of 1, 2, 5.5 and 11 Mbps, all 802.11g payload data rates of 6, 9, 12, 18, 24, 36, 48 and 54 Mbps, as well as 802.11n MCS0~MCS7, HT20/HT40, 800ns and 400ns guard interval.

It includes a single band WLAN CMOS efficient power amplifier (PA) and an internal low noise amplifier (LNA). The Radio Frequency Front-end is single-ended bi-directional input and output.

The SV6166F has additional LDOs and DCDC buck convertor that could provide noise isolation for digital and analog supplies and excellent power efficient with minimum BOM cost.

The SV6166F integrates a 2M byte (2MB)/ 16M bit (16Mb) flash in package. The SV6166F provides multiple peripheral interfaces including SPI_ MASTER, UART_DATA, UART_DEBUG, I2C_MASTER, I2S, etc.

The only external clock source needed for SV6166F based designs is a high speed crystal or oscillator. A variety of reference clocks are supported which include 19.2, 20, 24, 25, 26, 38.4, 40 and 52 MHz

SV6166F Features

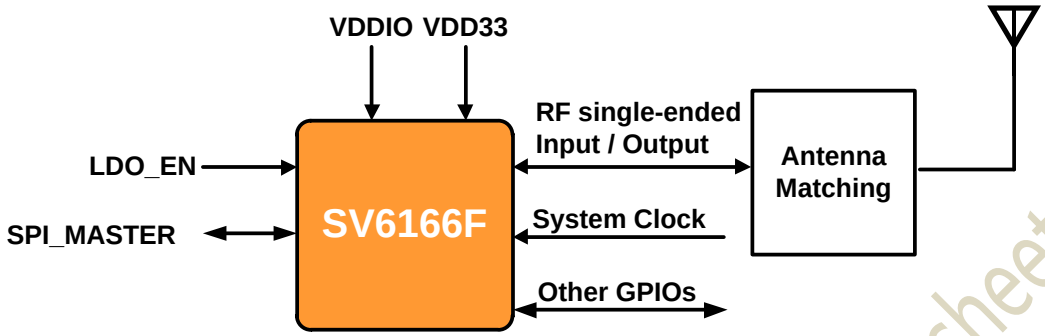
- All CMOS IEEE 802.11 b/g/n single chip
- Single stream 802.11n provides highest throughput and superior RF performance for embedded system
- Advanced 1x1 802.11n features:
  - Full / Half Guard Interval
  - Frame Aggregation
  - Reduced Inter-frame Space (RIFS)
  - Space Time Block Coding (STBC)
  - Greenfield mode
- Integrated WLAN CMOS efficient power amplifier with internal power detector and closed loop power calibration

ORDERING INFORMATION

Part Number	Package
SV6166F	Green/RoHS Compliant QFN 48L, 6x6 mm, 0.4mm pitch

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SV6166F System Block Diagram

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Revision History

Version	Date	Owner	Description
0.1	2017/10/20	WM / Hoz / Thomas	Initial draft
0.2	2018/03/26	Thomas	Add LDO current consumption
0.3	2018/06/06	Hoz/Thomas	Fixed GPIO7 strapping typo and LDO_EN timing typo
0.4	2018/07/30	Hoz Lin	Modify the description about flash

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Table of Contents

1 System Overview 5
2 Power Supplies and Power Management 8
3 DC Characteristics 12
4 Frequency References 16
5 Electrical Specifications 17
6 System Power Consumption 19
7 Pin Descriptions 20
8 Package Information 32

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1 SYSTEM OVERVIEW

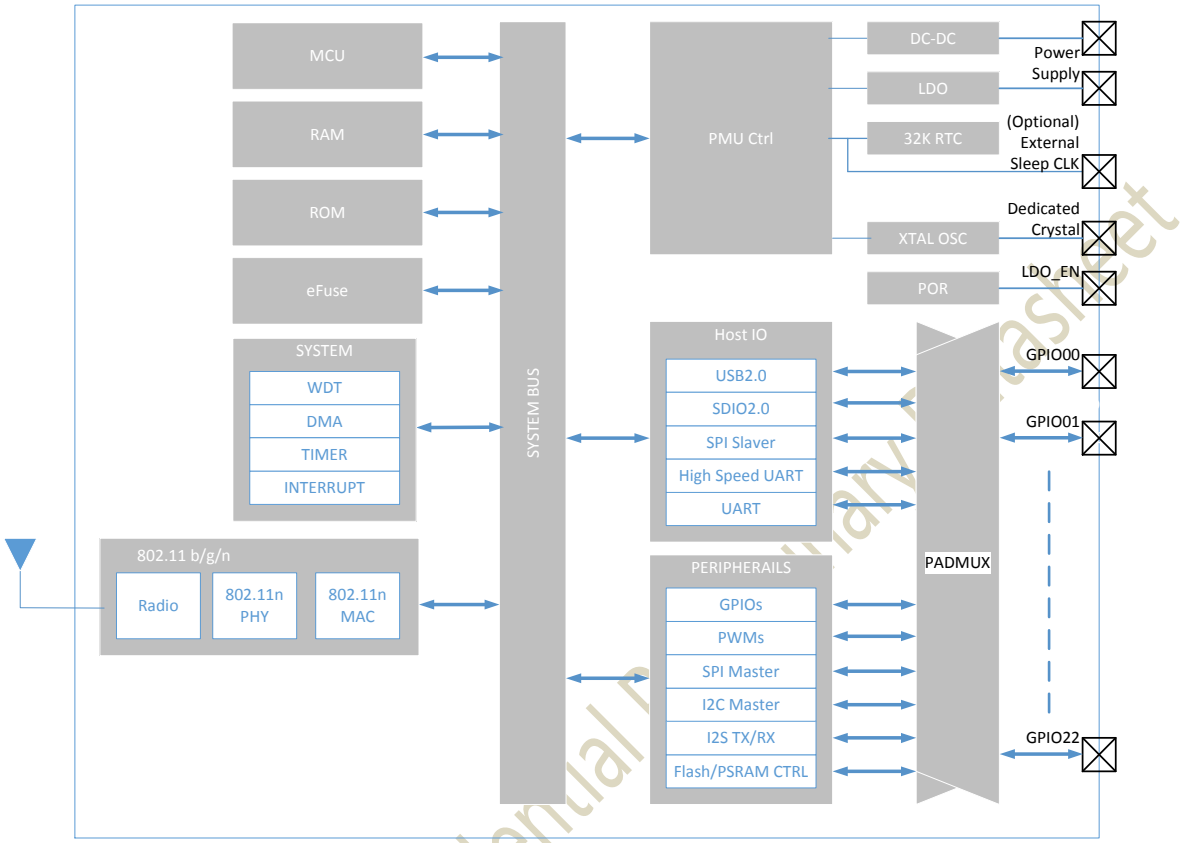


Figure 1-1: SV6166F Block Diagram

1.1 General Description

The SV6166F WLAN is designed to support IEEE 802.11 b/g/n single stream with the state of-the-art design techniques and process technology to achieve low power consumption and high throughput performance to address the requirement of mobile and handheld devices. The SV6166F WLAN low power function uses the innovative design techniques and the optimized architecture which best utilizes the advanced process technology to reduce active and idle power, and achieve extreme low power consumption at sleep state to extend the battery life. The SV6166F WLAN A-MPDU Tx function maximizes the throughput performance while achieving the best buffer utilization.

1.2 MAC Features

- 802.11 b/g/n/e/i/d
- WLAN/BT coexistence mechanisms
- 802.11n features
  - A-MPDU Tx & Rx
  - Support immediate Block-Ack

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- AP/STA mode
  - Soft-AP
- Rate adaption mechanism
- WFA features
  - WEP/TKIP/WPA/WPA2
  - WMM/WMM PS

1.3 PHY features

- 802.11b, 11g, and 802.11n 1T1R
- Short Guard Interval
- Greenfield mode
- RIFS in RX mode
- STBC in RX mode
- Enhanced and robust sensitivity for wider coverage range
- Supports calibration algorithm to handle no-idealities effects from CMOS RF block

1.4 SYSTEM

- Andes Technology N10 processor w/ ILM/DLM and I-cache.
- 128K ROM and 192 KB SRAM for Instruction and data SRAM in total.
- 8K retention SRAM.
- Integrate 2M byte (2MB)/ 16M bit (16Mb) SPI flash in package
- Suspend/Wake-up manger controller.
- Two channel DMA off load CPU loading.
- One Flash controller supports both Flash and PSRAM up to 16MB/64Mb with XIP
- One I2S TX/RX channel for 8~32bits/8~192KHz in master/slave mode.
- One SPI master
- One I2C master
- Five PWMs
- Four millisecond timers
- Four microsecond timers
- Two watchdog
- All pins can be multiplexed to GPIO by user scenario

1.5 HOST INTERFACE

- USB 2.0
- SDIO 2.0
  - Clock rate up to 50MHz
- SPI slave
  - Provide proprietary/user CMD
- High Speed UART
  - Support RTSN/CTSN/RX/TX, 4 pins
  - Baud rate up to 4.8MHz
- UART
  - Support RX/TX, 2 pins
  - Baud rate up to 921600

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1.6 System Clocking and Reset

The SV6166F has a system clocking block and reset which controls the clocks and power going to other internal modules. Its inputs consist of sleep requests from these modules and its outputs consist of clock enable and power signals which are used to gate the clocks going to internal modules. The system clocking and reset block also manages resets going to other modules within the device.

1.7 Design for Test

It also has features which enable testing of digital blocks via ATPG scan, memories via MBIST, analog components, and the radio.

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2 POWER SUPPLIES AND POWER MANAGEMENT

2.1 General Description and PMU Power Connection

The power management unit (PMU) contains Under-Voltage Lockout (UVLO) circuit, Low Dropout Regulators (LDOs), buck DC-DC converter and reference bandgap circuit.

The PMU integrated multi-LDOs and one buck converter. Those circuits are optimized for the given functions by balancing quiescent current, dropout voltage, line / load regulation, ripple rejection and output noise.

The input voltage of the buck converter is 3.3V. Its output voltage is 1.6V and feeds into the input power of the RF circuit and DLDO which has 1.2V output voltage for all digital circuits.

Figure 2-1 shows the typical power connection for SV6166F. DLDO and some RF circuits are powered by the buck converter output. The VDDIO is a power input which may be 1.8V or 3.3V from the host side. The connection structure is shown in the figure below.

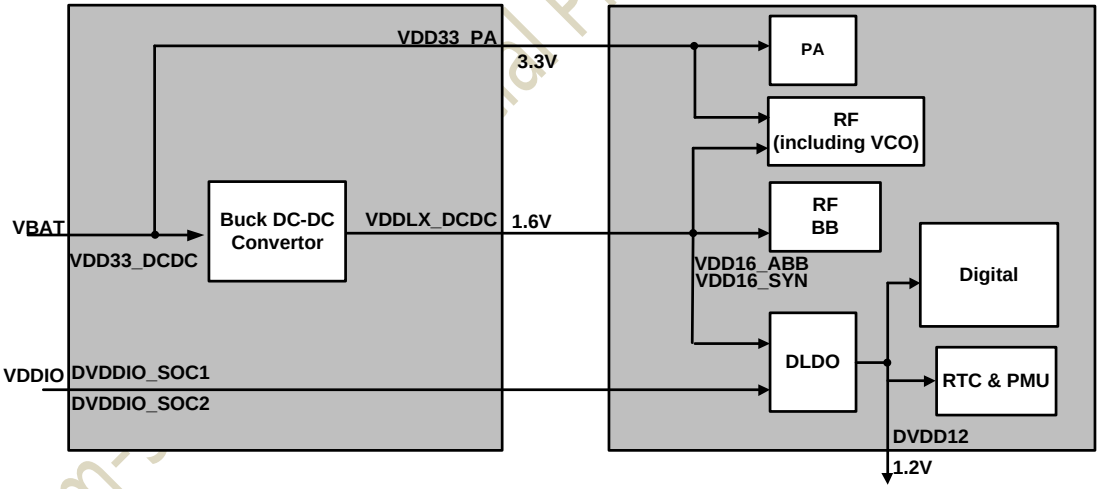


Figure 2-1: SV6166F Power Connection

2.2 Under-Voltage Lockout (UVLO)

The UVLO state in the PMU prevents startup if the initial voltage of the battery is below pre-defined threshold. It ensures that SV6166F is powered on with the battery in good condition. In addition, when the battery voltage is getting lower, it will enter the UVLO state, and the PMU will be turned off by itself to prevent further discharging.

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2.3 DLDO

The DLDO is integrated in the PMU to supply digital core. It converts voltage from 1.6V input to 1.2V output which suits the digital circuits. The input is typically connected to the buck's output.

2.4 Buck Converter

The regulator is a DC-DC step-down converter (buck converter) to source 300mA (max.) with 2.0V to 1.5V programmable output voltage based on the register setting. It supplies power for the RF circuit and DLDO.

2.5 Power Management Control

There are three power modes that SV6166F operates when it is initialized: HOST_OFF, ACTIVE mode and SLEEP mode. There are two intermediate system transition modes: FW_DOWNLOAD and WARM_UP mode. The following are the brief introduction to each mode.

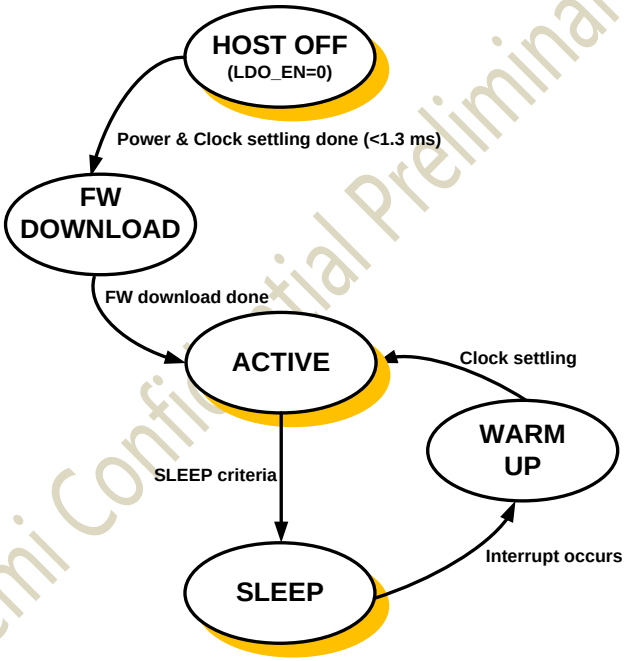


Figure 2-2: SV6166F Power State

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Table 2-1: SV6166F Power State Description

State	Description
HOST OFF	When **LDO_EN** pin is de-asserted and logically low, the chip is brought to this state immediately.
	Sleep clock and internal power supply is disabled.
	After **LDO_EN** pin is asserted, the internal power and clock will be settled down within 1.3 ms.
FW DOWNLOAD	States for firmware download after power and clock is settled down.
SLEEP	The host controller can determine when to enter sleep to turn off most circuit in SV6166F. All the RF, DPLL circuits are turned off. In sleep mode, the system could be awakened after the sleep time is expired or by an external wake up signal from the host controller.
	All internal states are maintained and the Crystal oscillator is disabled.
WARM UP	The system transitions from SLEEP to ACTIVE. The crystal or oscillator is brought up and the PLL is enabled.
ACTIVE	The high speed clock is operational and sent to each block by the clock control register.
	The RF circuit is enabled to transmit or receive data, and the whole system is under normal operation.

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2.6 Power-on Sequence

Figure 2-3 shows the power-on sequence of the SV6166F from power-up to firmware download, including the initial device power-on reset evoked by LDO_EN signal. The LDO_EN input level must be kept the same as VDDIO voltage level. After initial power-on, the LDO_EN signal can be held low to turn off the SV6166F or pulsed low to induce a subsequent reset. After LDO_EN is assert and host starts the power-on sequence of the SV6166F. From that point, the typical SV6166F power-on sequence is shown below:

1. Within 1.3 millisecond, the internal power-on reset (POR) will be done. And host could download firmware code of DPLL setting if the crystal is not default setting, 26MHz. The internal running clock is crystal frequency.
2. After 100us of DPLL settling time, host could set internal clock to full speed and finish all the downloading of firmware code.

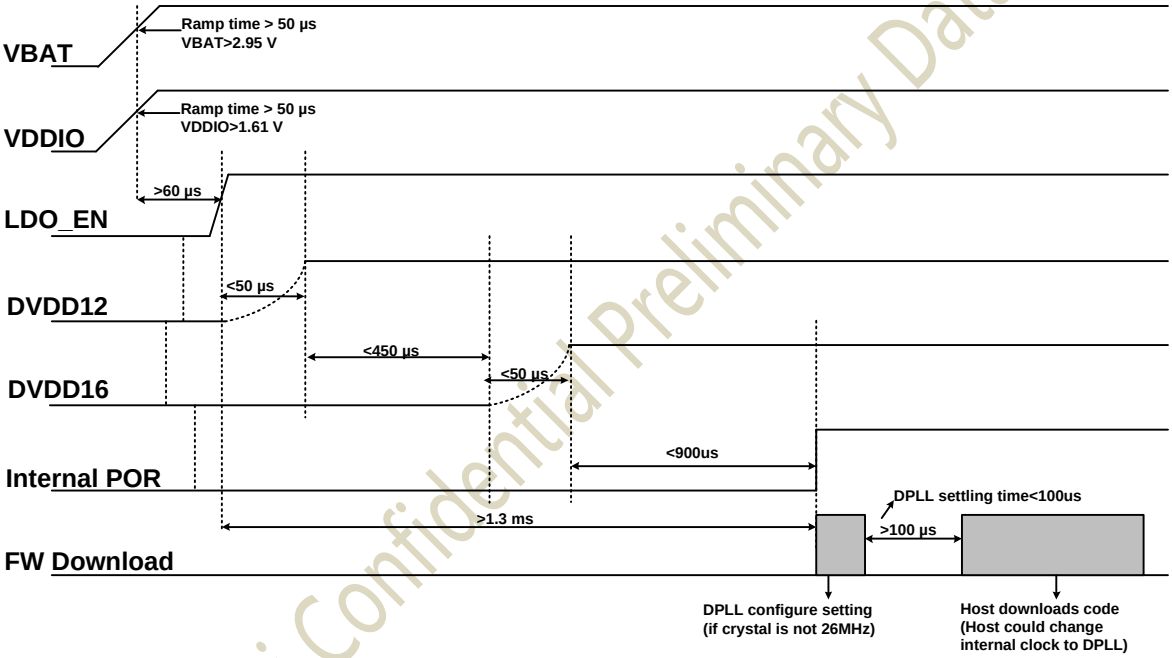


Figure 2-3: Power-on sequence

2.7 Reset Control

The SV6166F **LDO_EN** pin can be used to completely reset the entire chip. After this signal has been de-asserted, the SV6166F is in off mode waits for host communication. Until then, the MAC, BB, and SOC blocks are powered off and all modules are held in reset. Once the host has initiated communication, the SV6166F turns on its crystal and later on DPLL. After all clocks are stable and running, the resets to all blocks are automatically de-asserted.

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3 DC CHARACTERISTICS

3.1 Absolute Maximum Ratings

The absolute maximum ratings in Table 3-1 indicate levels where permanent damage to the device can occur, even if these limits are exceeded for only a brief duration. Functional operation is not guaranteed under these conditions. Operation at absolute maximum conditions for extended periods can adversely affect long-term reliability of the device.

Table 3-1: Absolute Maximum Ratings

Symbol (domain)	Description	Max Rating	Unit
VDD16	VDD input for analog 1.6V	-0.3 to 3.6	V
VDD33_SX	VDD input for external components I/O control	-0.3 to 3.6	V
VDD33_SX_5G	VDD input for external components I/O control	-0.3 to 3.6	V
VDD33_RF	VDD input for external components I/O control	-0.3 to 3.6	V
DVDDIO1	VDD input for GPIO pins	-0.3 to 3.6	V
DVDDIO2	VDD input for GPIO pins (same level as DVDDIO1)	-0.3 to 3.6	V
DVDD12	VDD output for internal digital circuit	-0.3 to 1.32	V
VDD16_DCDC	VDD input for digital circuit's LDO	-0.3 to 3.6	V
VDD33_DCDC	VDD input for DCDC	-0.3 to 3.6	V
VDD33_USB	VDD input for USB	-0.3 to 3.6	V
VDD33_SX	VDD input for external components I/O control	-0.3 to 3.6	V
VDD33_SX_5G	VDD input for external components I/O control	-0.3 to 3.6	V

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3.2 Environmental Ratings

The environmental ratings are shown in Table 3-2

Table 3-2 Environmental Ratings

Characteristic	Conditions/Comments	Value	Units
Ambient Temperature (T_A)	Functional operation	-20 to +85	˚C

3.2.1 Storage Condition

The calculated shelf life in sealed bag is 12 months if stored between 0°C and 40°C at less than 90% relative humidity (RH). After the bag is opened, devices that are subjected to solder reflow or other high temperature processes must be handled in the following manner:

a) Mounted within 168-hours of factory conditions < 30 **°C** /60%RH

b) Storage humidity needs to maintained at <10% RH

c) Baking is necessary if customer exposes the component to air over 168 hrs, baking condition: 125°C / 8hrs

3.3 Thermal Characteristics

Table 3-3: the thermal characteristics of the SV6166F

Thermal characteristics without external heat sink in still air condition

Symbol	Description	Typ.	Unit
T_J	Maximum Junction Temperature (Plastic Package)	125	°C
θ_{JA}	Thermal Resistance θ_{JA} (°C /W) for JEDEC 4L system PCB	37.8	°C/W
θ_{JC}	Thermal Resistance θ_{JC} (°C /W) for JEDEC 4L system PCB	TBD	°C/W
Ψ_{Jt}	Thermal Characterization parameter Ψ_{Jt} (°C /W) for JEDEC 4L system PCB	4.13	°C/W
	Maximum Lead Temperature (Soldering 10s)	260	°C

Notes: * JEDEC 51-7 system FR4 PCB size: 3" x 4.5" (76.2 x 114.3 mm)

* Thermal characteristics without external heat sink in still air condition

3.4 PMU Under Voltage Lock-out (UVLO) Characteristics

Table 3-4 PMU UVLO characteristics

Symbol (domain)	Description	Min.	Typ.	Max.	Unit
Under Voltage Lock-Out (UVLO)					
Under voltage rising threshold of VBAT	VDD33: pin VDD33_DCDC & VDD33_RF		2.95		V
Under voltage falling threshold of VBAT	VDD33: pin VDD33_DCDC & VDD33_RF		2.85		V
Under voltage rising threshold of DVDDIO	DVDDIO: pin DVDDIO_SOC1 & DVDDIO_SOC2		1.6		V
Under voltage falling threshold of DVDDIO	DVDDIO: pin DVDDIO_SOC1 & DVDDIO_SOC2		1.5		V

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3.5 Electrostatic Discharge Specifications

This is an ESD sensitive product! Observe precaution and handle with care. Extreme caution must be exercised to prevent electrostatic discharge (ESD) damage. Proper use of wrist and heel grounding straps to discharge static electricity is required when handling these devices.

Table 3-5: ESD Specifications

Pin Type	Test Condition	ESD Rating	Unit
Human Body Mode (HBM)	refers to MIL-STD-883G Method 3015.7	Pass ±2.5	KV
Charged Device Model(CDM)	JEDEC –500 +500 V specification JESD22-C101, all pins	Pass ±500	V

3.6 Recommended Operating Conditions and DC Characteristics

Table 3-6: Recommended Operating Conditions and DC Characteristics

Domain (Symbol)	Description	Min.	Typ.	Max.	Unit
VDD16	VDD input for analog 1.6V		1.6		V
VDD33_SX	VDD input for external components I/O control	2.1	3.3	3.46	V
VDD33_SX_5G	VDD input for external components I/O control	2.1	3.3	3.46	V
VDD33_RF	VDD input for external components I/O control	2.1	3.3	3.46	V
DVDDIO1	VDD input for GPIO pins	1.75	3.3	3.46	V
DVDDIO2	VDD input for GPIO pins (same level as DVDDIO1)	1.75	3.3	3.46	V
DVDD12	VDD output for internal digital circuit		1.2		V
VDD16_DCDC	VDD input for digital circuit's LDO		1.6		V
VDD33_DCDC	VDD input for DCDC	2.1	3.3	3.46	V
VDD33_USB	VDD input for USB	3.13	3.3	3.46	V
(V_{IL})	Input Low voltage when VDDIO=3.3V	-0.3		0.8	V
(V_{IH})	Input High voltage when VDDIO=3.3V	2		3.6	V
(V_{T+})	Schmitt trigger low to high threshold voltage when VDDIO=3.3V	1.6	1.74	1.89	V
(V_{T-})	Schmitt trigger high to low threshold voltage when VDDIO=3.3V	1.27	1.4	1.56	V

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(V_{OL})	Output low voltage when VDDIO=3.3V			0.4	V
(V_{OH})	Output high voltage when VDDIO=3.3V	2.4			V
(R_{PD})	Input weakly pull-down resistance when VDDIO=3.3V. All GPIO pins have internal weakly pull- down option except that GPIO_5 has internal weakly pull-up option				KΩ
(R_{PU})	Input weakly pull-high resistance when VDDIO=3.3V. All GPIO pins have internal weakly pull- down option except that GPIO_5 has internal weakly pull-up option				KΩ
(I_{OL})	Low level output current @ V_{OL}(max), 8mA setting	11.9	17.7	23.4	mA
	Low level output current @ V_{OL}(max), 12mA setting	15.8	23.5	31.1	mA
(I_{OH})	High level output current @ V_{OH}(min), 8mA setting	17.2	34.1	58.8	mA
	High level output current @ V_{OH}(min), 12mA setting	23.9	47.2	81.5	mA

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4 FREQUENCY REFERENCES

4.1 Crystal Oscillator Specifications

Table 4-1: Crystal Oscillator Specifications

Parameter	Condition/Notes	Min.	Typ.	Max.	Unit
Frequency Range	–	Between 19.2MHz~52MHz			
Crystal load Capacitance	–	–	10		pF
ESR	–	–	–	70	Ω
Frequency tolerance Initial and over temperature	–	-20ppm	–	20ppm	ppm

4.2 External Clock-Requirements and Performance

Table 4-2: External Clock-Requirements and Performance

Parameter	Condition/Notes	Min.	Typ.	Max.	Unit
Frequency Range	–	Between 19.2MHz~52MHz			
OSCIN Input Voltage	AC-couple analog signal	400	–	1500	mV_{PP}
Frequency tolerance Initial and over temperature	–	-20ppm	–	20ppm	ppm
Duty Cycle	26MHz clock	40	50	60	%
Phase Noise (802.11b/g)	26MHz clock at 1KHz offset	–	–	–119	dBc/Hz
	26MHz clock at 10KHz offset	–	–	–129	dBc/Hz
	26MHz clock at 100KHz offset	–	–	–134	dBc/Hz
	26MHz clock at 1MHz offset	–	–	–139	dBc/Hz
Phase Noise (802.11n 2.4GHz)	26MHz clock at 1KHz offset	–	–	–125	dBc/Hz
	26MHz clock at 10KHz offset	–	–	–135	dBc/Hz
	26MHz clock at 100KHz offset	–	–	–140	dBc/Hz
	26MHz clock at 1MHz offset	–	–	–145	dBc/Hz

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5 Electrical Specifications

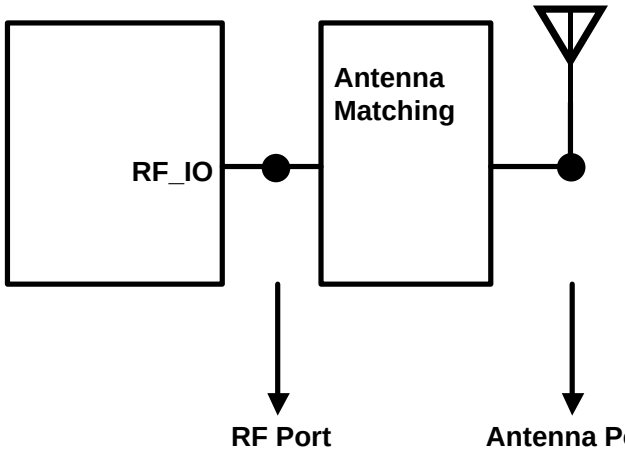


Figure 5-1: RF Front-End Reference Topology for RF Performance

Note: All specifications are measured at the Antenna Port unless otherwise specified.

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5.1 WLAN RF Performance Specifications

Table 5-1: WLAN RF Performance Specifications

Parameter	Condition/Notes	Min.	Typ.	Max.	Unit
Frequency Range		2412	-	2484	MHz
Rx Sensitivity (CCK)	CCK, 1 Mbps		-95.5		dBm
	CCK, 2 Mbps		-93.5		dBm
	CCK, 5.5 Mbps		-91.0		dBm
	CCK, 11 Mbps		-88.0		dBm
Rx Sensitivity (OFDM)	OFDM, 6 Mbps		-91.5		dBm
	OFDM, 9 Mbps		-90.0		dBm
	OFDM, 12 Mbps		-88.0		dBm
	OFDM, 18 Mbps		-86.0		dBm
	OFDM, 24 Mbps		-82.5		dBm
	OFDM, 36 Mbps		-79.5		dBm
	OFDM, 48 Mbps		-74.5		dBm
	OFDM, 54 Mbps		-73.5		dBm
Rx Sensitivity (HT20) Greenfield 800nS GI Non-STBC	HT20, MCS0		-91.0		dBm
	HT20, MCS1		-88.0		dBm
	HT20, MCS2		-86.0		dBm
	HT20, MCS3		-81.5		dBm
	HT20, MCS4		-79.0		dBm
	HT20, MCS5		-74.5		dBm
	HT20, MCS6		-73.5		dBm
	HT20, MCS7		-72.5		dBm
Parameter	Condition/Notes	Min.	Typ.	Max.	Unit
RX Adjacent Channel Rejection (CCK)	CCK, 1 Mbps (30 MHz offset)		41		dB
	CCK, 11 Mbps (25 MHz offset)		41		dB
RX Adjacent Channel Rejection (OFDM)	OFDM, 6 Mbps (25 MHz offset)		39		dB
	OFDM, 54 Mbps (25 MHz offset)		23		dB
RX Adjacent Channel Rejection (HT20)	HT20, MCS0 (25 MHz offset)		38		dB
	HT20, MCS7 (25 MHz offset)		21		dB
TX Output Power (with PADPD)	CCK, 1-11 Mbps		19		dBm
	OFDM, 54 Mbps		16		dBm
	HT20, MCS7		15		dBm

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6 System Power Consumption

Note: All results are measured at the condition that VIO and VBAT are 3.3V.

Table 6-1: Power Consumption at DCDC mode (DCDC buck convertor is enable)

WLAN Operational Modes	Typ.	Unit
OFF[a]	2	uA
Rx, CCK, 1 Mbps	75	mA
Rx, OFDM, 54 Mbps	82	mA
Rx, HT20, MCS7	82	mA
Rx, HT40, MCS7	90	mA
Tx, CCK, 1 Mbps	297	mA
Tx, 11B, 11 Mbps@18dBm	330	mA
Tx, OFDM, 54 Mbps	280	mA
Tx, HT20, MCS7	283	mA
Tx, HT40, MCS7	287	mA
Power-saving(MCU_on)[b] , DTIM1	TBD	mA
Power-saving(MCU_on)[b] , DTIM3	TBD	mA
Power-saving(MCU_off)[c] , DTIM1	TBD	mA
Power-saving(MCU_off)[c] , DTIM3	TBD	mA

Table 6-2: Power Consumption at LDO mode (DCDC buck convertor is disable)

WLAN Operational Modes	Typ.	Unit
OFF[a]	2	uA
Rx, CCK, 1 Mbps	106	mA
Rx, CCK, 11 Mbps	113	mA
Rx, OFDM, 54 Mbps	126	mA
Rx, HT20, MCS7	126	mA
Rx, HT40, MCS7	144	mA
Tx, CCK, 1 Mbps@18dBm	344	mA
Tx, OFDM, 54 Mbps@15dBm	311	mA
Tx, HT20, MCS7@15dBm	313	mA
Tx, HT40, MCS7@14dBm	307	mA
Power-saving(MCU_on)[b] , DTIM1	TBD	mA
Power-saving(MCU_on)[b] , DTIM3	TBD	mA
Power-saving(MCU_off)[c] , DTIM1	TBD	mA
Power-saving(MCU_off)[c] , DTIM3	TBD	mA

a. OFF mode test condition: VBAT=3.3V, VIO=3.3V, LDO_EN=0V (excluding Flash/PSRAM).

b. Intra-beacon Sleep when MCU is turn on.
It is used in the applications that require the CPU to be working.

c. Intra-beacon Sleep when MCU is turn off.

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7 Pin Descriptions

This section contains a listing of the signal descriptions (see Figure 7-1 for the SV6166F QFN package pin-out)

The following nomenclature is used for signal names:

NC	No connection should be made to this pin
P	At the end of the signal name, indicates the positive side of a differential signal
N	At the end of the signal name, indicates the negative side of a differential signal

The following nomenclature is used for signal types described in Table 6-1:

IA	Analog input signal
I	Digital input signal
IO	Digital bidirectional signal
IOA	Analog bidirectional signal
O	Digital output signal
P	Power signal
G	Ground signal

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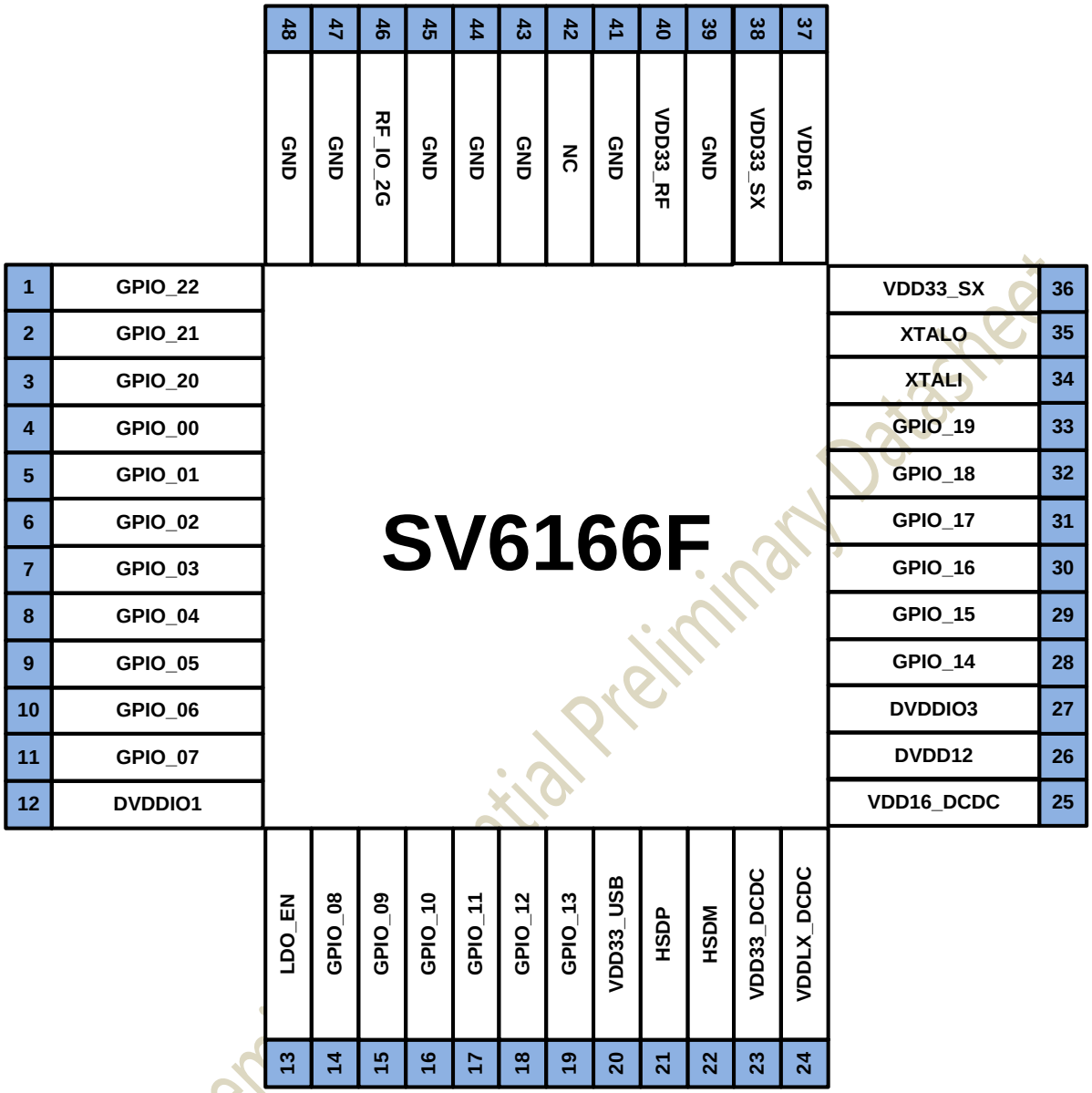


Figure 7-1: SV6166F QFN Pin Assignment (top view)

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Table 7-1: SV6166F Package Pin-out

No.	Name	Description	Type (default)
1	GPIO22	General Purpose I/O Pins	
2	GPIO21	General Purpose I/O Pins	
3	GPIO20	Strapping Purpose I/O Pins	
4	GPIO00	General Purpose I/O Pins	
5	GPIO01	General Purpose I/O Pins	
6	GPIO02	General Purpose I/O Pins	
7	GPIO03	General Purpose I/O Pins	
8	GPIO04	General Purpose I/O Pins	
9	GPIO05	General Purpose I/O Pins	
10	GPIO06	General Purpose I/O Pins	
11	GPIO07	Strapping Purpose I/O Pins	
12	DVDDIO1	VIO input 1	
13	LDO_EN	Reset signal to power down IC	
14	GPIO08	General Purpose I/O Pins	
15	GPIO09	General Purpose I/O Pins	
16	GPIO10	General Purpose I/O Pins	
17	GPIO11	General Purpose I/O Pins	
18	GPIO12	General Purpose I/O Pins	
19	GPIO13	General Purpose I/O Pins	
20	VDD33_USB	USB 3.3V power supply	
21	HSDP	USB DP pin	
22	HSDM	USB DM pin	
23	VDD33_DCDC	analog 3.3V input for DCDC	
24	VDDLX_DCDC	DCDC buck regulator: output to inductor	
25	VDD16_DCDC	DCDC 1.6V	
26	DVDD12	Digital 1.2V input	
27	DVDDIO3	VIO input 3	
28	GPIO14	Strapping Purpose I/O Pins	
29	GPIO15	Strapping Purpose I/O Pins	
30	GPIO16	General Purpose I/O Pins	
31	GPIO17	General Purpose I/O Pins	
32	GPIO18	General Purpose I/O Pins	
33	GPIO19	General Purpose I/O Pins	
34	XTALI	Input of crystal clock reference	
35	XTALO	Output of crystal clock reference	
36	VDD33_SX	analog 3.3V input	
37	VDD16	analog 1.6V input	
38	VDD33_SX	analog 3.3V input	
39	GND	Ground	
40	VDD33_RF	analog 3.3V input	
41	GND	Ground	
42	NC	NC	

iComm Semi Confidential Preliminary Datasheet

43	GND	Ground	
44	GND	Ground	
45	GND	Ground	
46	RF_IO	2.4 GHz RF input & output port	
47	NC	NC Pin	
48	GND	Ground	

7.1 Interface Selection

SV6166F has four bootstrap pins to select interface which including debug interface, host I/O and USB crystal frequency. There are GPIO07, GPIO14, GPIO15 and GPIO20 (see Table 7.1-1 for detail). The bootstrap pin of GPIO07 decides switching the debug interface (ICE or SPI DEBUG slave) out to GPIOs (see Table 7.1-2 for detail). The bootstrap pins of GPIO14 and GPIO15 decode switch the host IO (Flash/PSRAM, SDIO, SPI DATA slave or USB) out to GPIOs (see Table 7.1-3/4/5/6 for detail). The bootstrap pin of GPIO20 only provides the information of external crystal frequency for USB interface using.

Table 7.1-1: The strapping truth table

strapping truth table		
	Interface mode	Description
GPIO[07]		
0	ICE debug	for ICE debug interface
1	SPI debug	for SPI debug interface (default)
GPIO[15], GPIO[14]		
00	SPIFL-E	SPI flash w/l in-place execution (default)
01	SPIDATA	SPI data mode
10	SDIO	SDIO mode
11	USB	USB mode
GPIO[20]		
0	USB_XO25M	for USB 25MHz clock (default)
1	USB_XO40M	for USB 40MHz clock

Preliminary Datasheet

Table 7.1-2: The strapping pin of GPIO07

GPIO07		Low			High		
Pin No.	Name	I/O	PULL	I/O Function	I/O	PULL	I/O Function
4	**GPIO00**	I/O	F	TMSC	I/O	F	GPIO00
5	**GPIO01**	I	F	TCKC	I	F	SPI1_S_CLK
6	**GPIO02**	I	PU	nSRST	I	PU	SPI1_S_CSN
7	**GPIO03**	I	F	UART0_RXD	I	F	SPI1_S_MOSI
8	**GPIO04**	O	F	UART0_TXD	O	F	SPI1_S_MISO

Table 7.1-3: The strapping pin of GPIO14/15 for Flash/PSRAM interface

GPIO15/14		2'b00		
Pin No.	Name	I/O	PULL	I/O Function
9	**GPIO05**	O	F	UART1_RTS
10	**GPIO06**	I	F	UART1_RXD
11	**GPIO07**	O	PU	UART1_TXD
14	**GPIO08**	I/O	F	GPIO08
15	**GPIO09**	I	F	UART1_CTS
16	**GPIO10**	I/O	F	GPIO10
17	**GPIO11**	I/O	F	GPIO11
18	**GPIO12**	I/O	F	GPIO12
19	**GPIO13**	O	PU	SPI_PSRAM_CSN
28	**GPIO14**	I/O	PD	SPI_flash_IO0_DI (MOSI)
29	**GPIO15**	O	PD	SPI_flash_CLK
30	**GPIO16**	I/O	PU	SPI_flash_IO3_HD
31	**GPIO17**	O	PU	SPI_flash_CSN
32	**GPIO18**	I/O	PD	SPI_flash_IO1_DO (MISO)
33	**GPIO19**	I/O	PU	SPI_flash_IO2_WP
3	**GPIO20**	I/O	F	GPIO20
2	**GPIO21**	I	F	I2C_S_SCL
1	**GPIO22**	I/O	F	I2C_S_SDA

iComm Confidential Preliminary Datasheet

Table 7.1-4: The strapping pin of GPIO14/15 for SPI DATA slave interface

GPIO15/14		2'b01		
Pin No.	**Name**	**I/O**	**PULL**	**I/O Function**
9	**GPIO05**	O	F	UART1_RTS
10	**GPIO06**	I	F	UART1_RXD
11	**GPIO07**	O	PU	UART1_TXD
14	**GPIO08**	O	F	SPI0_S_INT(GPIO09)
15	**GPIO09**	I	F	UART1_CTS
16	**GPIO10**	I	F	SPI0_S_MOSI
17	**GPIO11**	I	F	SPI0_S_CLK
18	**GPIO12**	O	F	SPI0_S_MISO
19	**GPIO13**	I	F	SPI0_S_CSN
28	**GPIO14**	I/O	F	GPIO14
29	**GPIO15**	I/O	F	GPIO15
30	**GPIO16**	I/O	F	GPIO16
31	**GPIO17**	I/O	F	GPIO17
32	**GPIO18**	I/O	F	GPIO18
33	**GPIO19**	I/O	F	GPIO19
3	**GPIO20**	O	F	WIFI_WAKE_HOST(GPIO20)
2	**GPIO21**	I	F	HOST_WAKE_WIFI(GPIO21)
1	**GPIO22**	O	F	WIFI_CLK_REQ(RTC_RDY_V12D)

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Table 7.1-5: The strapping pin of GPIO14/15 for SDIO2.0 interface

GPIO15/14		2'b10		
Pin No.	**Name**	**I/O**	**PULL**	**I/O Function**
9	**GPIO05**	I/O	F	GPIO05
10	**GPIO06**	I	F	UART1_RXD
11	**GPIO07**	O	PU	UART1_TXD
14	**GPIO08**	I/O	F	SD_D2
15	**GPIO09**	I/O	F	SD_D3
16	**GPIO10**	I/O	F	SD_CMD
17	**GPIO11**	I	F	SD_CLK
18	**GPIO12**	I/O	F	SD_D0
19	**GPIO13**	I/O	F	SD_D1
28	**GPIO14**	I/O	F	GPIO14
29	**GPIO15**	I/O	F	GPIO15
30	**GPIO16**	I/O	F	GPIO16
31	**GPIO17**	I/O	F	GPIO17
32	**GPIO18**	I/O	F	GPIO18
33	**GPIO19**	I/O	F	GPIO19
3	**GPIO20**	O	F	WIFI_WAKE_HOST(GPIO20)
2	**GPIO21**	I	F	HOST_WAKE_WIFI(GPIO21)
1	**GPIO22**	O	F	WIFI_CLK_REQ(RTC_RDY_V12D)

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Table 7.1-6: The strapping pin of GPIO14/15 for USB2.0 interface

GPIO15/14		2'b11		
Pin No.	**Name**	**I/O**	**PULL**	**I/O Function**
9	**GPIO05**	O	F	UART1_RTS
10	**GPIO06**	I	F	UART1_RXD
11	**GPIO07**	O	PU	UART1_TXD
14	**GPIO08**	I/O	F	GPIO08
15	**GPIO09**	I	F	UART1_CTS
16	**GPIO10**	I/O	F	GPIO10
17	**GPIO11**	I/O	F	GPIO11
18	**GPIO12**	I/O	F	GPIO12
19	**GPIO13**	I/O	F	GPIO13
28	**GPIO14**	I/O	F	GPIO14
29	**GPIO15**	I/O	F	GPIO15
30	**GPIO16**	I/O	F	GPIO16
31	**GPIO17**	I/O	F	GPIO17
32	**GPIO18**	I/O	F	GPIO18
33	**GPIO19**	I/O	F	GPIO19
3	**GPIO20**	O	F	WIFI_WAKE_HOST(GPIO20)
2	**GPIO21**	I	F	HOST_WAKE_WIFI(GPIO21)
1	**GPIO22**	O	F	WIFI_CLK_REQ(RTC_RDY_V12D)

Icomm-semi Confidential Preliminary Datasheet

7.2 User Define I/O Function Selection

After bootstrap, the SV6166F also provides a pad multiplex switching from the bootstrap function to selected I/O function by register signals. There is a condition to leave bootstrap function. That is switching to GPIO first then switching to select I/O function. The table 7.2-1 shows the all I/O functions for each PAD.

Table 7.2-1: The PAD multiplex for each PAD

Pin No.	Pin Name	Alternate Functions		
4	GPIO00	strapping function		
		I/O	U	GPIO[00]
		O	F	gpo_int
		O	F	PWM0
		O	F	WIFI_TX_SW
5	GPIO01	strapping function		
		I/O	U	GPIO[01]
		O	F	gpo_int
		O	F	PWM1
		O	F	WIFI_RX_SW
6	GPIO02	strapping function		
		I/O	U	GPIO[02]
		O	F	gpo_int
		O	F	PWM2
		O	F	BT_SW
7	GPIO03	strapping function		
		I/O	U	GPIO[03]
		O	F	gpo_int
		O	F	PWM3
		I	F	BT_PRIORITY
		I	F	UART0_RXD
8	GPIO04	strapping function		
		I/O	U	GPIO[04]
		O	F	gpo_int
		O	F	PWM4
		O	F	WLAN_ACTIVE
		O	F	UART0_TXD
9	GPIO05	strapping function		
		I/O	U	GPIO[05]

Icomm-semi Confidential Preliminary Datasheet

		O	F	gpo_int
		I/O	F	I2S_BCLK
10	GPIO06	strapping function		
		I/O	U	GPIO[06]
		O	F	gpo_int
		I	F	I2S_DI
11	GPIO07	strapping function		
		I/O	U	GPIO[07]
		O	F	gpo_int
		O	F	I2S_DO
14	GPIO08	strapping function		
		I/O	U	GPIO[08]
		O	F	gpo_int
		I/O	F	I2S_BCLK
		O	L	SPI_M_CLK
15	GPIO09	strapping function		
		I/O	U	GPIO[09]
		O	F	gpo_int
		I/O	F	I2S_LRCLK
16	GPIO10	strapping function		
		I/O	U	GPIO[10]
		O	F	gpo_int
		I	F	I2S_DI
		I	H	SPI_M_MISO
		I/O	H	I2C_M_SDA
17	GPIO11	strapping function		
		I/O	U	GPIO[11]
		O	F	gpo_int
18	GPIO12	strapping function		
		I/O	U	GPIO[12]
		O	F	gpo_int
		O	F	I2S_DO
		O	H	SPI_M_MOSI
		O	H	I2C_M_SCL
19	GPIO13	strapping function		
		I/O	U	GPIO[13]
		O	F	gpo_int
		I/O	F	I2S_LRCLK

Icomm-semi Confidential Preliminary Datasheet

		O	H	SPI_M_CS
		O	PU	SPI_PSRAM_CSN
28	GPIO14	strapping function		
		I/O	U	GPIO[14]
		O	F	gpo_int
		I/O	PD	SPI_flash_IO0_DI
29	GPIO15	strapping function		
		I/O	U	GPIO[15]
		O	F	gpo_int
		O	PD	SPI_flash_CLK
30	GPIO16	strapping function		
		I/O	U	GPIO[16]
		O	F	gpo_int
		I/O	PU	SPI_flash_IO3_HD
		O	F	SPI1_S_MISO
31	GPIO17	strapping function		
		I/O	U	GPIO[17]
		O	F	gpo_int
		O	PU	SPI_flash_CSN
32	GPIO18	strapping function		
		I/O	U	GPIO[18]
		O	F	gpo_int
		I/O	PD	SPI_flash_IO1_DO
		I	F	SPI1_S_CSN
33	GPIO19	strapping function		
		I/O	U	GPIO[19]
		O	F	gpo_int
		I/O	PU	SPI_flash_IO2_WP
		I	F	SPI1_S_MOSI
3	GPIO20	strapping function		
		I/O	U	GPIO[20]
		O	F	gpo_int
		O	F	I2S_MCLK
		O	F	I2S_MCLK
		I	F	SPI1_S_CLK
2	GPIO21	strapping function		
		I/O	U	GPIO[21]
		O	F	gpo_int

Icomm-semi Confidential Preliminary Datasheet

		O	H	I2C_M_SCL
		I	F	I2C_S_SCL
		O	F	UART0_TXD
1	GPIO22	strapping function		
		I/O	U	GPIO[22]
		O	F	gpo_int
		I/O	H	I2C_M_SDA
		I/O	F	I2C_S_SDA
		I	F	UART0_RXD
		I	F	BT_ACTIVE

Icomm-semi Confidential Preliminary Datasheet

8 PACKAGE INFORMATION

6 x 6 mm (body size), 0.4mm pitch QFN-48

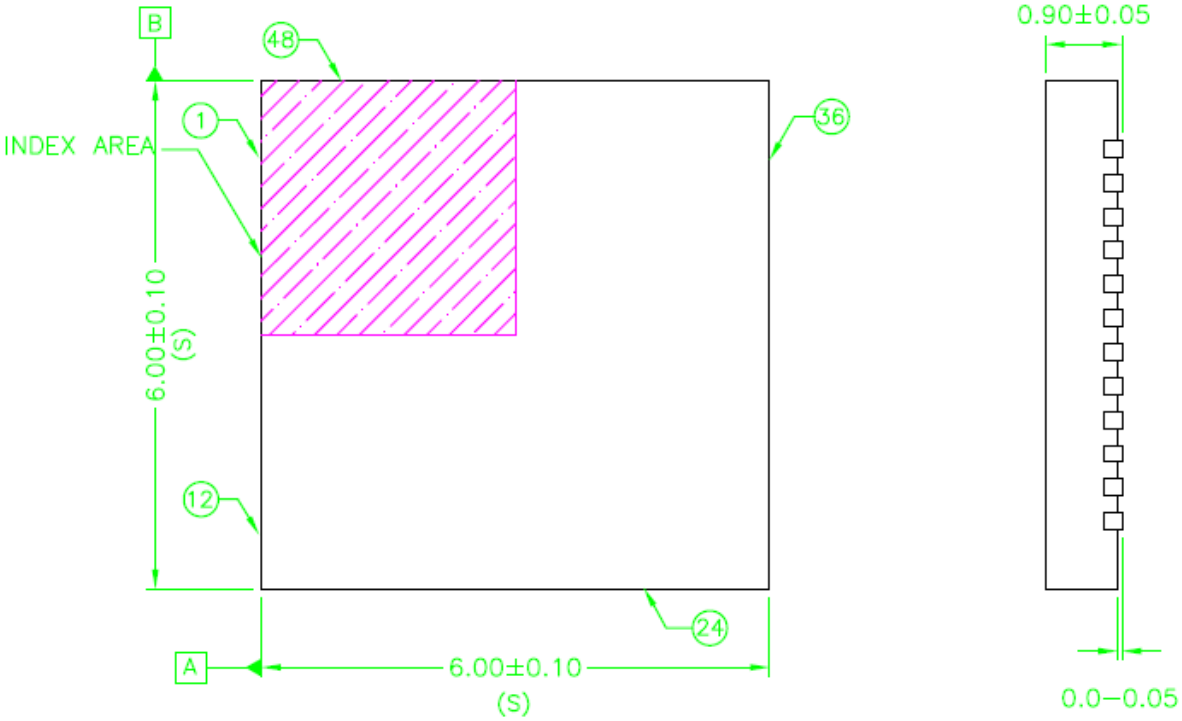
Marking format (top view)

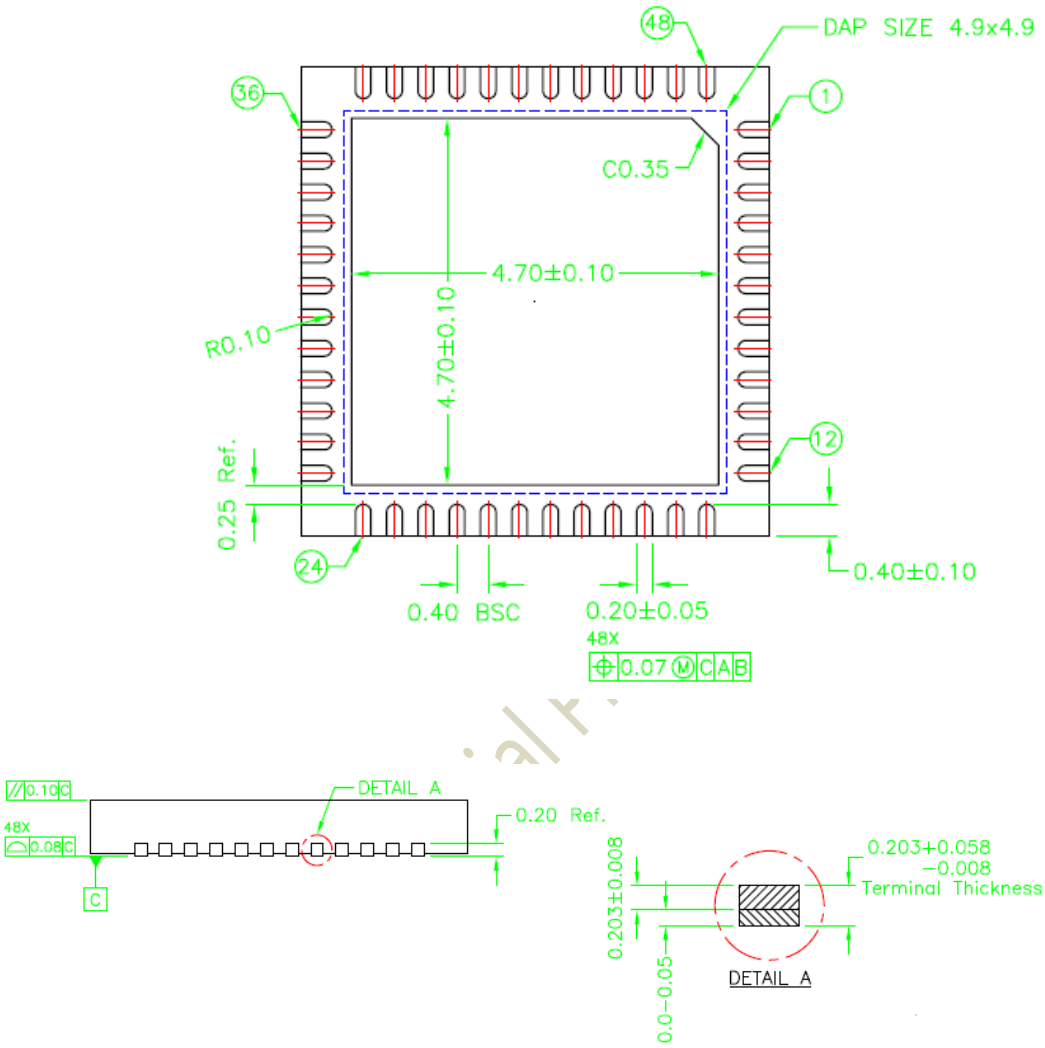


Pin #48

Indicates Pin #1(Laser Marked)

Preliminary Datasheet





NOTE :

1. ALL DIMENSION ARE IN mm. ANGLES IN DEGREES.
2. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS. COPLANARITY SHALL NOT EXCEED 0.08 mm.
3. WARPAGE SHALL NOT EXCEED 0.10 mm.
4. PACKAGE LENGTH / PACKAGE WIDTH ARE CONSIDERED AS SPECIAL CHARACTERISTIC. (S)
5. L/F STOCK# FR9012 (PPF)

Figure 8-1: SV6166F QFN 6 x 6 mm Package Dimensions